

PA²Perturb: Prompt-Activated Amplified Perturbation against Unauthorized Topological Dataset Usage

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Abstract—Graph-structured datasets can be leveraged to pre-train representationally powerful graph neural networks (GNNs). However, with the growing demand for high-quality graph data, the utilization of unauthorized datasets for training GNN models is becoming increasingly common, which seriously damages the interests of the dataset owners. Current ownership-preserving research has focused on validating unauthorized usage of unstructured datasets through non-intrusive and intrusive methods. Non-intrusive methods rely on the inherent decision boundaries that fail to guarantee the reliability of the results. Intrusive methods utilize perturbations to validate dataset ownership but easily damage the performance of the model due to intentional manipulations. To address these issues, we propose a novel prompt-activated amplified perturbation from the perspective of utility-preserving injection, called PA²Perturb. Specifically, to improve the stealthiness of perturbations, we utilize the cue-sensitive imperceptible subgraph to replace anomalous modifications, which is subject to semantic and distributional similarity constraints. Subsequently, to maximally preserve the utility of the dataset and model, our scheme does not modify node labels and only utilizes prompts to activate latent patterns and manipulate model behaviors. Eventually, to enhance the transferability of our scheme, we utilize prompts to adaptively amplify perturbations to accommodate different scenarios. Overall, the dataset owner is allowed to check the usage of datasets by observing the behavioral characteristics of models during prompt-activated queries to amplify perturbations. Comprehensive experiments on various representative graph datasets demonstrate that our PA²Perturb outperforms existing state-of-the-art baselines with verification accuracy over 95.0%. Our source code is publicly available at <https://anonymous.4open.science/r/PA2Perturb-4F14>.

Index Terms—Article submission, IEEE, IEEEtran, journal, LATEX, paper, template, typesetting.

I. INTRODUCTION

IN recent years, graph neural networks (GNNs) have achieved remarkable performance across diverse applications, including social network analysis [1]–[5], molecular property prediction [6]–[9], and financial risk assessment [10]–[12]. These advances have been largely driven by the availability of quality graph datasets [13]–[15], which capture rich structural and semantic relationships. However, with the growing demand for high-quality graph data, the utilization of unauthorized datasets to train GNN models is becoming increasingly common. A particularly noteworthy example is that over 50 million Facebook user records were improperly collected to train user profile models and used for political profiling without informed consent [16]. Dataset stealers extract data to train GNN models without the knowledge of dataset

owners, which poses copyright infringement issues. Accordingly, the unauthorized use of datasets seriously infringes upon the legitimate rights and interests of dataset owners, who have invested significant time and funds in data collection and cleaning.

Existing research [17]–[27] on dataset ownership verification (DOV) can be classified into two categories: non-intrusive methods and intrusive methods. Non-intrusive methods utilize the inherent characteristics of the dataset as fingerprints [24]–[30] to verify the ownership of the dataset without modifying the original dataset. Unfortunately, non-intrusive methods have obvious shortcomings in terms of stability and stealthiness. Specifically, non-intrusive methods are vulnerable to adversarial perturbations since they typically depend on statistical features as fingerprints. In addition, non-intrusive methods generally lack sufficient reliability and durability due to not modifying the data itself, which makes it difficult to provide verifiable and tamper-resistant evidence of ownership. Considering the complex structure of graph data and the lability of the unique features, non-intrusive methods fail to achieve effective ownership verification.

In contrast, intrusive methods [18], [19], [21]–[23], [31]–[33] involve embedding specific triggers to verify dataset ownership. Specifically, the dataset owner deliberately modifies part of the dataset to embed specific behavioral patterns, which causes unauthorized models to inadvertently learn these clues. Compared with non-intrusive methods, intrusive methods generally have higher verification accuracy and robustness, which are currently the main framework for verifying dataset ownership. Unfortunately, previous methods have three major shortcomings: 1) the embedded triggers suffer from limited stealthiness and stability, since they typically rely on abnormal modifications that lead to significant distributional shifts; 2) models trained on perturbed datasets show inferior adaptability in graph domains, and task-invariant structural alterations affect the generalization ability of the network on scenarios with different perturbation sensitivities; 3) these methods tend to cause irreversible degradation of topological datasets and trained architectures, while message-passing mechanisms in graph-structured data propagate and amplify local anomalous manipulations to damage predictive performance.

To effectively adapt upstream semantics to various downstream applications, advanced graph prompt learning (GPL) paradigms are introduced with time-efficiency and structure-awareness. In this way, the cross-domain perturbations could remain valid with adaptation via such prompting technology. Motivated by this understanding, our key insight is to utilize activated prompts to amplify imperceptible perturbations, which achieves adaptive and transferable dataset ownership

This paper was produced by the IEEE Publication Technology Group. They are in Piscataway, NJ.

Manuscript received April 19, 2021; revised August 16, 2021.

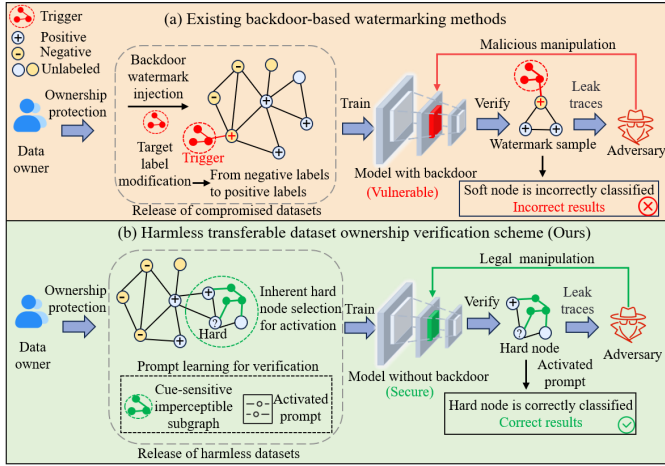


Fig. 1. Comparison between existing backdoor-based watermarking methods and our harmless transferable dataset ownership verification scheme.

verification. However, due to the complexity and disturbance sensitivity of topological datasets, there are three challenges in judging ownership of graph domains: 1) stealthy perturbation in tightly-coupled graph data poses fundamental difficulties, in which even minor modifications propagate via message passing to induce global distribution shifts; 2) model transferability under injected perturbations remains non-trivial to preserve, particularly when fixed perturbations affect training objectives and alter decision boundaries across various tasks; 3) reliable verification without irreversible damage to the dataset and model is complex to achieve, as perturbations should be powerful enough to enable ownership identification without performance degradation and loss of data usability.

To address the above shortcomings and challenges, we present a novel prompt-activated amplified perturbation from the perspective of utility-preserving injection, named PA²Perturb, which consists of the cue-sensitive imperceptible subgraph (CIS) and the activated prompt. Our scheme integrates the distributional and semantic features of the dataset into CIS, while the model is guided by prompts to generate different responses to perturbed nodes. Specifically, to improve the stealthiness and stability of perturbations, we align perturbations with original datasets in terms of semantic similarity and distributional similarity, then smoothly inject them into stable regions in the graph. Subsequently, to maximally preserve the utility of the dataset and model, we only inject the imperceptible subgraph on hard nodes (i.e., easily misclassified nodes) and utilize prompts to correct the prediction of models. Eventually, to improve the transferability of our scheme, we utilize the activated prompt to align with soft node (i.e., reliably classifiable nodes) and hard node features, and adaptively amplify perturbations to achieve transferable ownership verification. As shown in Fig. 1(a), backdoor-based methods memorize the implicit correspondence between the perturbation and the target label. In graph data, the message passing mechanism will amplify local feature patterns, which leads to a decline in overall model performance. In contrast, our scheme can verify the usage of datasets by analyzing the behavior on perturbed nodes. This involves

injecting imperceptible subgraphs into the original graph and constructing activated prompts to achieve utility-preserving dataset ownership verification, as illustrated in Fig. 1(b). In summary, the contributions of our work are summarized as follows:

- We, for the first time, propose a novel cross-architecture graph-structured dataset ownership verification scheme, which integrates semantic and structural features to generate imperceptible subgraphs and utilizes activated prompts to amplify perturbations. As a result, the scheme enables effective and transferable dataset ownership verification.
- The CIS leverages semantic and distributional characteristics to generate imperceptible perturbations injected into stable regions for ownership verification. In addition, we utilize prompts to identify different nodes and adaptively amplify perturbations to ensure the transferability across various prevailing GNN models and graph datasets.
- Comprehensive experiments on various representative graph datasets demonstrate that our PA²Perturb outperforms existing methods with the verification confidence over 95.0%. Under various attack methods, our scheme still maintains a high accuracy rate above 90.6% compared with 87.3% of competitive baselines on Flickr. In addition, PA²Perturb also shows excellent transferability across various popular GNN models and maintains accuracy consistently above 83.7% on Flickr.

II. RELATED WORK

A. Graph Prompt Learning

Prompt learning, initially successful in large language models (LLMs), has been extended to the graph domain. Compared to the parameter-laborious fine-tuning, in the GPL [34]–[41], light-weight prompts are tuned to bridge the gap between pretrained GNNs and downstream tasks. Existing GPL methods follow two principal paradigms: prompt as token and prompt as graph. The prompt as token paradigm injects learnable prompt vectors into intermediate layers of the GNN to modulate message passing. For instance, GraphPrompt [35] augments the readout phase with task-specific prompt tokens, which enables flexible adaptation to diverse downstream objectives. In contrast, the prompt as graph paradigm appends a learnable subgraph directly to the input graph. ProG [34] is a representative method. It treats the graph prompt as a virtual structure that conditions the encoder without modifying its parameters. Unlike previous GPL methods primarily designed for the performance of a specific task, our scheme leverages prompts to adaptively amplify perturbations and enables effective ownership verification of graph datasets.

B. Topology-Aware Trigger Embedding

Most existing trigger embedding methods [42], [43] in graph domains rely on static modification patterns, which overlook the inherent semantic and structural characteristics of the target nodes. To address this limitation, recent research [44]–[48] has undertaken the exploration of topology-aware trigger embedding, in which triggers are embedded in combination

with the semantic and the distribution knowledge of the graph. For instance, Dai et al. [46] achieved semantically similar embeddings through a process that increases the feature similarity between the trigger and the attached node. Zhang et al. [45] constrained the generated triggers to be consistent with the distribution of the original dataset from an out-of-distribution perspective for robust embeddings. However, these triggers are still associated with the target class, which inevitably damages the dataset and model performance. Unlike prior methods, we inject perturbations that are jointly constrained by both semantics and distributional similarity without association with specific labels. This makes it possible to facilitate smooth injection and avoid damage to the dataset and model.

C. Dataset Ownership Verification

Traditional dataset protection methods, which are primarily aimed at preventing unauthorized access and ensuring data privacy, constitute a foundational and pivotal area of research. Key techniques in this domain include encryption [49], [50], digital watermarking [18], [19], [21], [22], [51]–[54], and privacy-preserving approaches [20], [55]. Dataset ownership verification has emerged as a significant topic within the realm of dataset protection. To the best of our knowledge, existing dataset ownership verification methods can be mainly divided into non-intrusive methods [24]–[27], [56], [57] and intrusive methods [18], [19], [21]–[23], [58]. Non-intrusive methods utilize the inherent characteristics of datasets to verify ownership. For instance, Maini et al. [27] proposed MinGD, which evaluated the dataset ownership based on the distance between the data and the decision boundary. Chen et al. [59] leveraged loss-based membership signals to construct fingerprint features for ownership verification. However, non-intrusive methods rely on the inherent characteristics of datasets for verification, which suffer from vague verification boundaries and limited transferability. In contrast, the key insight of the intrusive methods is to inject specific triggered perturbations into datasets that are significantly correlated with the target classification. While recent studies [18], [19], [21]–[23] have verified the use of datasets by injecting triggered perturbation, these methods typically use fixed forms of injection perturbation, which makes it difficult to transfer between different tasks and models. In this paper, we propose a novel intrusive scheme for ownership verification, which has underlying cross-semantic transferability driven by activated prompts.

III. PRELIMINARIES ANALYSIS

A. Notations

We denote an undirected graph as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, X)$, where $\mathcal{V} = \{v_1, \dots, v_N\}$ represents the set of N nodes, $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ denotes the set of edges, and $X = \{x_1, \dots, x_N\}$ is the set of node attributes, with each $x_i \in \mathbb{R}^d$ representing the feature vector of node v_i . The adjacency matrix of the graph $\mathcal{G}$ is denoted by $A \in \mathbb{R}^{N \times N}$, where $A_{ij} = 1$ if nodes v_i and v_j are connected, and $A_{ij} = 0$ otherwise. The one-hop neighborhood of node v_i , including the node itself, is denoted by $\mathcal{N}_i$.

Considering the widespread real-world applications of graph-related tasks and their representativeness in existing research, we select a semi-supervised node classification setting, where only a small subset of nodes $\mathcal{V}_L \subset \mathcal{V}$ in the training graph $\mathcal{G}_T$ are labeled with ground-truth labels from a predefined label space $\mathcal{L} = \{1, \dots, C\}$. The remaining nodes $\mathcal{V}_U = \mathcal{V} \setminus \mathcal{V}_L$ are unlabeled during training. The corresponding label set is denoted as $\mathcal{Y} = \{y_1, \dots, y_N\}$, where $\mathcal{Y}_L$ and $\mathcal{Y}_U$ refer to the labels of $\mathcal{V}_L$ and $\mathcal{V}_U$, respectively. In the inductive setting, we assume the existence of a separate graph $\mathcal{G}_T = (\mathcal{V}_T, \mathcal{E}_T, X_T)$, whose label information is not accessible during training. The goal is to learn a node classifier that generalizes well from $\mathcal{G}_T$ to unseen nodes or graphs, while it also makes use of both structural and attribute information from labeled nodes.

B. Graph Neural Networks

Given an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, X)$ with node features $\{\mathbf{x}_v\}_{v \in \mathcal{V}}$, an L -layer GNN encoder performs a two-step computation at each layer: message construction and state update. Specifically, for each node $v \in \mathcal{V}$ at layer l ($0 \leq l < L$), the process is defined as

$$\begin{aligned} \mathbf{a}_v^{(l)} &= \text{AGG} \left(\left\{ M^{(l)} \left(\mathbf{h}_u^{(l)}, \mathbf{h}_v^{(l)}, \mathbf{e}_{uv}; \boldsymbol{\theta}_M^{(l)} \right) \mid u \in \mathcal{N}(v) \right\} \right), \\ \mathbf{h}_v^{(l+1)} &= U^{(l)} \left(\mathbf{h}_v^{(l)}, \mathbf{a}_v^{(l)}; \boldsymbol{\theta}_U^{(l)} \right), \end{aligned} \quad (1)$$

where $\mathbf{h}_v^{(l)} \in \mathbb{R}^d$ denotes the representation of node v at layer l , $\mathbf{e}_{uv}$ is an optional edge feature, and $\mathcal{N}(v)$ denotes the set of neighbors of node v . $M^{(l)}$ represents the message function, $\text{AGG}(\cdot)$ is a permutation-invariant aggregation operator, and $U^{(l)}$ denotes the update function, typically implemented as a linear transformation followed by a non-linear activation. The learnable parameter sets $\boldsymbol{\theta}_M^{(l)}$ and $\boldsymbol{\theta}_U^{(l)}$ control the behavior of the message and update functions, respectively.

C. Graph Prompt Learning

Current GPL methods are divided into two paradigms: prompt as token, represented by GraphPrompt [35], and prompt as subgraph, represented by ProG [34]. Despite their structural differences, both frameworks recast pre-training and downstream inference into a unified graph-level task and employ learnable prompts to guide task-specific adaptation.

Prompt as subgraph. Structural prompts introduce a learnable subgraph $\mathcal{G}_p = (P, S)$ into the input graph. The node set $P = \{v_1^p, \dots, v_{|P|}^p\}$ contains $|P|$ prompt nodes, each associated with an independent, trainable embedding $\mathbf{p}_i \in \mathbb{R}^d$. The edge set $S \subseteq P \times P$ defines the internal topology of the prompt. By grafting $\mathcal{G}_p$ onto a target graph $\mathcal{G}$, one obtains an augmented graph $\tilde{\mathcal{G}} = \mathcal{G} \cup \mathcal{G}_p$, which is then processed by a frozen GNN encoder. During the few-shot adaptation, only the prompt embeddings $\{\mathbf{p}_i\}$ and the task head are updated, while the encoder parameters remain fixed. ProG is a representative implementation of this paradigm, and ProG-Meta [34] extends it with meta-learning for enhanced cross-task generalization.

Prompt as token. The token-based paradigm preserves the original graph structure and instead injects a small set of learnable vectors directly into the hidden layers of a frozen

GNN. For a downstream task j , let $\mathbf{p}_j \in \mathbb{R}^d$ denote the prompt token and $\mathbf{h}_v$ the layer- L embedding of node v . A re-weighted readout combines node embeddings and the token via element-wise multiplication:

$$\mathbf{z}_G^{(j)} = \text{Readout}\left(\{\mathbf{p}_j \odot \mathbf{h}_v \mid v \in \mathcal{V}\}\right), \quad (2)$$

where $\odot$ denotes the hadamard product and $\text{Readout}(\cdot)$ is a symmetric pooling function. Only $\mathbf{p}_j$ and the task head are optimized, while the backbone GNN remains unchanged. GraphPrompt demonstrates that a small number of tokens can effectively adapt large pretrained encoders to diverse objectives without modifying their internal parameters.

Unlike the prompt as subgraph that directly optimizes the entire dataset, GraphPrompt generates a unique weight for each node. This weight enables flexible control over the enhancement effect of the prompt on individual nodes. Accordingly, our scheme adopts the prompt as token paradigm. Specifically, the activated prompt is instantiated as a learnable vector and injected into every node embedding. This design enhances the representation capacity of the GNN, while it also enables dataset ownership verification.

IV. THREAT MODEL

Attacker's goals. The attacker aims to illegally exploit the graph dataset to train or fine-tune GNNs without authorization. The ultimate goal is to build a well-performing model that can evade any ownership verification mechanism deployed by the dataset owner. Specifically, the attacker seeks to retain high predictive performance on classification tasks and eliminate any embedded cues that may expose the unauthorized dataset usage during auditing.

Attacker's capability. We assume that the attacker has full access to the graph dataset but lacks knowledge of the ownership verification method, which includes the injected perturbations and the prompt design. The attacker can fully control the training process and model architecture. However, the attacker cannot directly observe activated prompts and perturbation strategies used in ownership verification method.

Attacker's knowledge. We categorize attackers into adaptive and non-adaptive based on their knowledge of the latent perturbations. The adaptive attacker is aware of the imperceptible perturbations and can adopt mitigation strategies to compromise the effectiveness of ownership verification. Consequently, the injected perturbations can be partially or completely removed from the perturbed dataset. In contrast, the non-adaptive attacker is unaware of the presence of perturbations and does not employ any targeted countermeasures.

Owner's goal and capability. We assume that the dataset owner has complete manipulation of the dataset. Dataset owners do not permit the unauthorized use of datasets to train models. Owners have the ability to train surrogate models to generate effective perturbations and prompts that verify ownership of the dataset. However, owners are unaware of architecture and parameters of suspected models, which only allow them to identify dataset ownership through suspected model outputs.

V. METHODOLOGY

A. Overview

In contrast to previous methods, we propose PA²Perturb treats dataset ownership verification as an adaptive amplified perturbation and prompt-activated verification problem. Our key insight is that carefully crafted adaptive prompts optimized on a surrogate model are consistent with the behavior patterns of models trained on the same perturbed dataset, where such consistency enables effective dataset ownership verification. Driven by this motivation, we utilize prompt-activated amplified perturbations to achieve a transferable dataset ownership verification. In general, this verification scheme consists of two parts: amplified perturbations and activated prompts.

Perturbation injection. Given an original graph dataset to be protected, as shown in Fig. 2(a), we identify a set of hard nodes through the surrogate model. These nodes are further expanded into context-aware stable regions through joint semantic and structural characteristics. We inject the CIS generated under the constraints of distribution and semantic similarity into the stable region. Moreover, we develop an adaptive prompt learning framework composed of soft-aware and hard-aware prompts that adaptively amplifies perturbations for dataset ownership verification and maintains model performance, as shown in Fig. 2(b). In practice, we adopt a bi-level optimization strategy to jointly optimize CIS and activated prompts so that they can converge collaboratively.

Perturbation verification. As illustrated in Fig. 2(c), to determine whether a suspected model violates ownership, the owner can observe whether the model exhibits similar behavioral characteristics to legal models on the perturbed dataset. The dataset owner injects CIS into the original dataset and utilizes adaptive activated prompts to amplify perturbations, which influences the behavioral characteristics of models trained on the perturbed dataset. To enable harmless dataset ownership verification, imperceptible perturbations are only injected to stable regions and activated via prompts, which does not affect the classification performance on soft nodes.

B. Context-Aware Stable Region Selection

In this subsection, we present the details of the region selection algorithm. Unlike the method of directly injecting perturbations into arbitrarily selected nodes, we select hard nodes and expand them into stable regions by integrating semantic and structural characteristics. These regions are carefully designed to achieve two core objectives: harmlessness and stability. The former ensures that perturbations within the selected region do not degrade the performance of the model, while the latter ensures that the region maintains structural integrity and consistent perturbation effects under different training conditions or graph operations. The resulting regions exhibit semantic consistency and structural smoothness, which makes them a key foundation for stealthy perturbations embedding.

Hard node selection. Given the extensive and diverse nature of graph data, the selection of perturbed nodes should adhere to two principles: 1) the selected nodes should effectively influence the classification results of the model; 2)

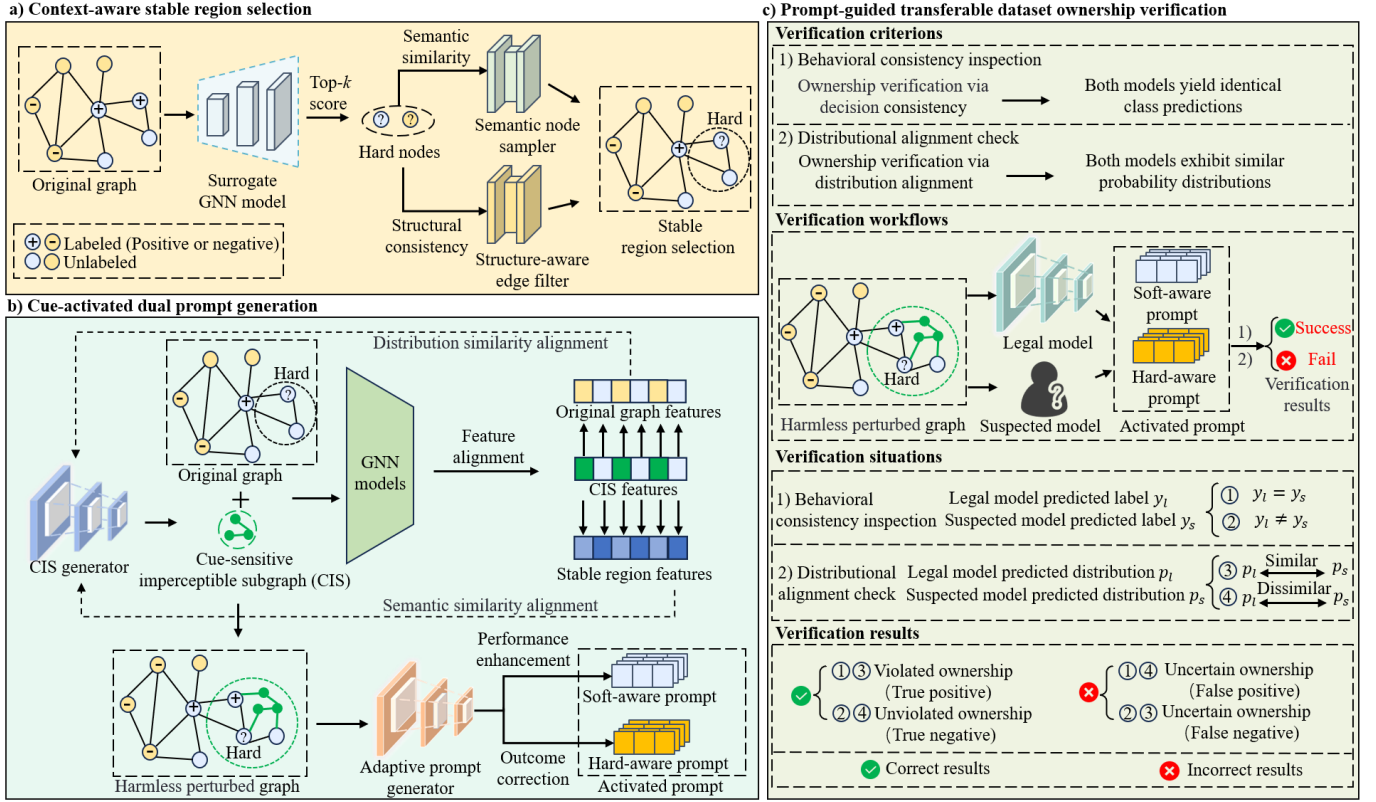


Fig. 2. The pipeline of PA²Perturb consists of three stages. (a) The context-aware region selection identifies hard nodes, which are then used to extract semantically and structurally stable regions for perturbation injection. (b) The activated prompt is trained to amplify CIS perturbations and influences model behavioral characteristics. (c) The ownership verification evaluates whether the suspected model is trained on the owner dataset by dual prompts.

the perturbation of these nodes should not adversely affect the performance of the classifier on clean dataset. Therefore, we select samples with high classification uncertainty in each category as perturbation nodes.

Given a labeled training set $\mathcal{V}_L \subset \mathcal{V}$, we first train a surrogate GNN classifier f_s on the original graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$:

$$\mathcal{L}_{\text{cls}} = - \sum_{v \in \mathcal{V}_L} \sum_{c=1}^C y_v^{(c)} \log f_s^{(c)}(\mathbf{x}_v), \quad (3)$$

where $y_v^{(c)} \in \{0, 1\}$ denotes the one-hot ground-truth label of node v , and $f_s^{(c)}(\cdot)$ is the predicted probability of class c from the surrogate model.

Once the surrogate classifier is trained, we compute prediction entropy and pseudo labels for each node $v \in \mathcal{V}_U$ in the unlabeled set:

$$H(v) = - \sum_{c=1}^C f_s^{(c)}(v) \cdot \log f_s^{(c)}(v), \quad (4)$$

$$\hat{y}_v = \arg \max_c f_s^{(c)}(v).$$

For each class $c \in \mathcal{C}$, we rank the nodes with pseudo label $\hat{y}_v = c$ by descending entropy and select the top- k most uncertain nodes to construct a balanced hard node set:

$$\mathcal{V}_h = \bigcup_{c=1}^C \text{TopK}_{v \in \mathcal{V}_U, \hat{y}_v = c} H(v). \quad (5)$$

Semantic node sampler. To ensure semantic coherence within each stable region, we perform neighborhood selection based on semantic similarity. For each hard node $v_h \in \mathcal{V}_h$, we extract its local K -hop subgraph $\mathcal{G}_h = (\mathcal{V}_h^{(K)}, \mathcal{E}_h)$. We denote the embedding of node v produced by the fixed surrogate encoder as $\mathbf{h}_v = f_s^{\text{emb}}(\mathbf{x}_v)$. A node $u \in \mathcal{V}_h^{(K)}$ is retained if its cosine similarity with the hard node exceeds a threshold τ_s . The semantically aligned node set is thus defined as:

$$\text{sim}(v_h, u) = \frac{\mathbf{h}_{v_h}^\top \mathbf{h}_u}{\|\mathbf{h}_{v_h}\| \cdot \|\mathbf{h}_u\|} \geq \tau_s, \quad (6)$$

$$\mathcal{V}_{\text{sem}}(v_h) = \left\{ u \in \mathcal{V}_h^{(K)} \mid \text{sim}(v_h, u) \geq \tau_s \right\}.$$

Structure-aware edge filter. To preserve the topological consistency of each region, we filter edges based on structural consistency. We define the relative degree difference between v_h and its neighbor u as:

$$\delta_d(v_h, u) = \left| \frac{\deg(v_h) - \deg(u)}{\deg(v_h)} \right|. \quad (7)$$

Edges that satisfy $\delta_d(v_h, u) \leq \tau_d$ are retained to form the structure aware edge set.

$$\mathcal{E}_{\text{str}}(v_h) = \{(v_h, u) \in \mathcal{E}_h \mid \delta_d(v_h, u) \leq \tau_d\}. \quad (8)$$

We then define the stable region centered at v_h and use the union of all such regions as the foundation for subsequent perturbation and verification modules:

$$\mathcal{R}(v_h) = (\mathcal{V}_{\text{sem}}(v_h), \mathcal{E}_{\text{str}}(v_h)). \quad (9)$$

C. Cue-Activated Dual Prompt Generation

To enable the activation and verification of stealth perturbations, we jointly optimize two functional modules: a cue-sensitive subgraph generator f_c , and an adaptive dual prompt generator f_p . The former injects imperceptible perturbation subgraphs, while the latter modulates node representations through learnable prompts to improve classification performance and amplify perturbations. These two modules are trained alternately under a bi-level optimization scheme.

CIS generator. To enhance the imperceptibility of perturbations, we align the embedding representations of the CIS with those of the original graph in terms of distributional similarity, and with the embeddings of the connected stable regions in terms of semantic similarity.

Given the stable region $\mathcal{R}_v$ for each selected hard node v , we generate a cue-sensitive subgraph $\mathcal{G}_{\text{cis}}$ with learnable node features. We extract node embeddings via a frozen GNN encoder f_s , and model them as gaussian distributions for both the CIS and original graphs. For a node set $\mathcal{S} \in \{\mathcal{V}_{\text{cis}}, \mathcal{V}_{\text{ori}}\}$, its empirical mean and covariance are computed as:

$$\mu_{\mathcal{S}} = \frac{1}{|\mathcal{S}|} \sum_{v \in \mathcal{S}} \mathbf{h}_v, \quad \Sigma_{\mathcal{S}} = \frac{1}{|\mathcal{S}|} \sum_{v \in \mathcal{S}} (\mathbf{h}_v - \mu_{\mathcal{S}})(\mathbf{h}_v - \mu_{\mathcal{S}})^\top. \quad (10)$$

To quantify the distributional discrepancy between the CIS and the original graph, we compute the squared 2-Wasserstein distance between the two gaussian distributions:

$$\mathcal{L}_{\text{dist}} = \|\mu_{\text{cis}} - \mu_{\text{ori}}\|_2^2 + \text{Tr} \left(\Sigma_{\text{cis}} + \Sigma_{\text{ori}} - 2 \left(\Sigma_{\text{ori}}^{1/2} \Sigma_{\text{cis}} \Sigma_{\text{ori}}^{1/2} \right)^{1/2} \right). \quad (11)$$

To preserve local semantic coherence, we align the CIS node embeddings with the centroid of their attached stable region:

$$\mathbf{c}_v = \frac{1}{|\mathcal{R}_v|} \sum_{u \in \mathcal{R}_v} \mathbf{h}_u, \quad \mathcal{L}_{\text{sem}} = \sum_{v \in \mathcal{V}_{\text{cis}}} \left(1 - \frac{\mathbf{h}_v^\top \mathbf{c}_v}{\|\mathbf{h}_v\|_2 \cdot \|\mathbf{c}_v\|_2} \right). \quad (12)$$

The CIS generator parameters are updated via gradient descent using the total loss $\mathcal{L}_c$, which ensures that each generated subgraph minimizes statistical and semantic deviations from the stable region support.

$$\mathcal{L}_c = \mathcal{L}_{\text{dist}} + \gamma \cdot \mathcal{L}_{\text{sem}}. \quad (13)$$

Adaptive prompt generator. To effectively calibrate the classifier across different node types, we design a dual prompt generator f_p that modulates node representations based on their hardness. Specifically, we insert the generated CIS into the original graph to obtain a perturbed graph $\tilde{\mathcal{G}}$, upon which the prompt training is performed. In this perturbed graph, we introduce a prompt matrix $\mathbf{P} = [\mathbf{P}_{\text{soft}}; \mathbf{P}_{\text{hard}}] \in \mathbb{R}^{2 \times d_p}$, where $\mathbf{P}_{\text{soft}}$ and $\mathbf{P}_{\text{hard}}$ are learnable embeddings for soft and hard nodes, respectively. We initialize $\mathbf{P}_{\text{soft}}$ and $\mathbf{P}_{\text{hard}}$ using the average embedding of the nodes classified as soft nodes and hard nodes, respectively. Then, the initial values for the prompt embeddings are set as:

$$\mathbf{P}_{\text{soft}} = \frac{1}{|\mathcal{V}_{\text{soft}}|} \sum_{v_i \in \mathcal{V}_{\text{soft}}} \mathbf{x}_i, \quad \mathbf{P}_{\text{hard}} = \frac{1}{|\mathcal{V}_{\text{hard}}|} \sum_{v_j \in \mathcal{V}_{\text{hard}}} \mathbf{x}_j, \quad (14)$$

where $\mathbf{x}_i$ denotes the feature vector of node v_i , and $|\cdot|$ denotes the cardinality of a set.

Each node v is assigned a binary selector $\pi_v \in \{[1, 0], [0, 1]\}$, which is determined by whether the node is soft or hard. The modulated node embedding is computed as:

$$\tilde{\mathbf{h}}_v = \mathbf{h}_v \odot (\pi_v^\top \mathbf{P}). \quad (15)$$

This formulation enables a unified and differentiable prompt assignment mechanism, which allows the prompt matrix to specialize representations across tasks in a lightweight and interpretable manner.

To train the prompt generator, we fix the backbone GNN classifier f_s and optimize only f_p by minimizing the following cross-entropy based prompt loss:

$$\mathcal{L}_{\text{prompt}} = \sum_{v \in \mathcal{V}_L} \mathcal{L}_{CE}(f_s(\tilde{\mathbf{h}}_v), y_v) + \beta \sum_{v \in \mathcal{V}_h} \mathcal{L}_{CE}(f_s(\tilde{\mathbf{h}}_v), \hat{y}_v), \quad (16)$$

where $\hat{y}_v$ denotes the pseudo label predicted by the frozen GNN for hard nodes, and β controls the relative importance of hard node correction.

Bi-level alternating optimization. To jointly optimize the CIS generator f_c and the dual prompt generator f_p , we adopt a bi-level alternating optimization. This design allows each module to benefit from the feedback of the other, which ensures that the perturbations are both stealthy and functional, while the prompts are adaptively tuned to expose model behavior.

We formulate the overall training as a constrained optimization problem, where the classifier $f_c(\cdot; \theta_c)$ minimizes the total CIS loss, subject to an optimal prompt module $f_p(\cdot; \theta_p^*)$, as follows:

$$\min_{\theta_c} \mathcal{L}_{\text{CIS}}(f_c(\cdot; \theta_c), f_p(\cdot; \theta_p^*))$$

$$\min_{\theta_c} \mathcal{L}_c(\theta_c, \theta_p^*) \quad \text{s.t.} \quad \theta_p^* = \arg \min_{\theta_p} \mathcal{L}_p(\theta_p, \theta_c). \quad (17)$$

Inner objective: Given a fixed perturbation structure from the CIS generator, the inner loop focuses on optimizing the prompt parameters θ_p . The goal is to maximize the downstream utility of prompts in both regular node classification and signal activation. The prompt parameters are updated via gradient descent:

$$\theta_p^{(t+1)} = \theta_p^{(t)} - \eta_p \cdot \nabla_{\theta_p} \mathcal{L}_p(\theta_p^{(t)}, \theta_c). \quad (18)$$

Outer objective: Once the prompt module reaches a stable state, we update the CIS generator by minimizing the total perturbation loss $\mathcal{L}_c$, which accounts for both distributional and semantic alignment. The updated CIS patterns are more likely to retain stealth and improve downstream prompt response:

$$\theta_c^{(k+1)} = \theta_c^{(k)} - \eta_c \cdot \nabla_{\theta_c} \mathcal{L}_c(\theta_c^{(k)}, \theta_p^*). \quad (19)$$

This alternating training process is repeated until convergence. Through this interplay, f_c learns to generate amplified perturbations, while f_p becomes increasingly effective at decoding these signals for reliable ownership verification. Additionally, we have provided a detailed bi-level alternating optimization process in Appendix A.

D. Prompt-Guided Transferable Dataset Ownership Verification

In this stage, we verify the dataset ownership through a comparison of the prediction behaviors of the suspected model and the legal model, when both are guided by a fixed dual prompt generator f_p . The core idea is to assess consistency on two dimensions: behavioral consistency and distributional alignment. We use a shared set of hard nodes $\mathcal{V}_h$ for this assessment. Ownership is confirmed only if both criteria are satisfied.

Hard node embedding with prompt amplification. For each hard node $v \in \mathcal{V}_h$, we first obtain its GNN embedding $\mathbf{h}_v = f_s(v)$, where f_s is the fixed encoder used during prompt training. The embedding is amplified by a learned prompt vector $\mathbf{P}$ to form a prompt-guided representation:

$$\tilde{\mathbf{h}}_v = \mathbf{h}_v \odot \mathbf{P}, \quad (20)$$

where $\odot$ denotes element-wise multiplication. The same prompt is applied to both the legal model f_s and the suspected model f'_s .

Behavioral consistency. Let y_v^L and y_v^S be the predicted labels of node $v \in \mathcal{V}_h$ from the legal and suspected models, respectively. Behavioral consistency is evaluated by the proportion of hard nodes with matched predictions:

$$\mathcal{A}_{\text{label}} = \frac{1}{|\mathcal{V}_h|} \sum_{v \in \mathcal{V}_h} \mathbb{I}(y_v^L = y_v^S), \quad (21)$$

where $\mathbb{I}(\cdot)$ is the indicator function. A high agreement ratio implies behavioral similarity, which suggests potential unauthorized use of the dataset.

Distributional consistency. Let $\mathbf{p}_v^L = f_s(\tilde{\mathbf{h}}_v)$ and $\mathbf{p}_v^S = f'_s(\tilde{\mathbf{h}}_v)$ be the softmax output distributions of node v in the two models. The empirical maximum mean discrepancy (MMD) between the two sets is:

$$\begin{aligned} \text{MMD}^2(\mathcal{P}^L, \mathcal{P}^S) &= \frac{1}{|\mathcal{V}_h|^2} \sum_{i,j} k(\mathbf{p}_i^L, \mathbf{p}_j^L) + \sum_{i,j} k(\mathbf{p}_i^S, \mathbf{p}_j^S) \\ &\quad - 2 \sum_{i,j} k(\mathbf{p}_i^L, \mathbf{p}_j^S), \end{aligned} \quad (22)$$

where $k(\cdot, \cdot)$ is the gaussian kernel. To assess statistical significance, a permutation test is conducted by randomly shuffling the labels of $\mathcal{P}^L \cup \mathcal{P}^S$ and computing MMD repeatedly. The resulting p-value is:

$$p = \frac{1}{T} \sum_{t=1}^T \mathbb{I}(\text{MMD}_t^2 \geq \text{MMD}_{\text{true}}^2), \quad (23)$$

where T is the total number of permutations. A small p indicates remarkable evidence of distributional divergence.

Ownership verification criteria. To establish dataset ownership, we evaluate two distinct but complementary criteria: behavioral consistency derived from hard node label predictions, and distributional alignment derived from predicted probability distributions.

Let $\hat{y}_v^{\text{leg}}$ and $\hat{y}_v^{\text{susp}}$ denote the predicted labels on hard node $v \in \mathcal{V}_h$ by the legal and suspected models, respectively. We

define the behavioral consistency for node v as satisfied if $\hat{y}_v^{\text{leg}} = \hat{y}_v^{\text{susp}}$. Let $\mathcal{A}_{\text{label}}$ denote the proportion of hard nodes where this condition holds.

For distributional alignment, we compute the MMD between the predicted distributions of both models on $\mathcal{V}_h$, followed by a permutation test to obtain a statistical p-value p . A smaller p-value (e.g., $p \leq \alpha$) implies remarkable evidence that the two distributions are significantly different. Based on these two indicators, the verification outcome is interpreted as follows:

- **Confirmed infringement.** When hard nodes receive the same predicted labels from both models, and the output distributions are also statistically similar (i.e., $p \leq \alpha$), we conclude that the suspected model has been trained on the perturbed graph.
- **No infringement.** If the models disagree on label predictions for the hard nodes, and their output distributions are also dissimilar (i.e., $p > \alpha$), the suspected model is deemed not to have used the perturbed graph.
- **False positive.** If label predictions are consistent across models, but the distributional similarity is low (i.e., $p > \alpha$), this suggests the suspected model may have generalized well despite not being exposed to the perturbed graph.
- **False negative.** If the models disagree on label predictions, but the predicted distributions remain close (i.e., $p \leq \alpha$), it could indicate that the number or stability of hard nodes is insufficient, which leads to weak perturbation.

E. Theoretical Foundation of $\text{PA}^2\text{Perturb}$

Theorem 1. The power of prompt-activated perturbation amplification. Under a first-order approximation of f_θ around prompt-modulated representations, the proposed dual-prompt mechanism satisfies the following for hard nodes v and for any two GNN architectures f_θ and $f_{\theta'}$ trained on the same graph:

$$\Delta f_\theta(v) \geq \zeta \|\delta\|_2 \|\pi_v^\top \mathbf{P}\|_\infty, \quad (24)$$

$$|\Delta f_\theta(v) - \Delta f_{\theta'}(v)| \leq \eta \cdot D_{\text{MMD}}(\mathcal{D}_\theta, \mathcal{D}_{\theta'}), \quad (25)$$

where $\zeta > 1$ is the amplification coefficient that selectively enhances responses of uncertain nodes, η is a transferability constant, and $D_{\text{MMD}}(\cdot, \cdot)$ denotes the maximum mean discrepancy between the output distributions of the two models. The full theoretical derivations are presented in Appendix C.

VI. EVALUATION

A. Datasets and GNN Models

We evaluate our proposed scheme on four real-world benchmarks: Cora, Citeseer, Pubmed [13], and Flickr [14]. These datasets have been extensively adopted in prior work on GNN security analysis [42], [43], [45]–[47]. Specifically, Cora, Citeseer, and Pubmed are small-scale citation networks, while Flickr is a large-scale graph constructed from image captions with shared attributes. Meanwhile, to evaluate the performance of our scheme on mixed datasets, we combine

TABLE I
GRAPH STATISTICS FOR GNN TRAINING.

Datasets	#Graphs	#Nodes	#Edges	#Attributes	#Classes
Cora	1	2,708	5,429	1,433	7
Citeseer	1	3,327	9,104	3,703	6
Pubmed	1	19,717	88,648	500	3
Flickr	1	89,250	899,756	500	7

Cora with Pubmed and Flickr respectively to construct two mixed datasets: Cora-P and Cora-F. A brief description of each dataset is provided below, and their statistical properties are summarized in Table I. Following the setting in prior work [43], [45], [46], we adopt node classification as the main task. Within the GPL framework, we employ four widely-used GNN models: GCN [60], GraphSAGE [15], GAT [61], and GIN [62].

B. Graph Prompt Learning

We evaluate model performance with representative GPL methods that are designed for various downstream tasks. In particular, we consider GraphPrompt [35] as a comparative method, and develop an enhanced variant named GPrompt-G for dataset ownership verification. Refer to Appendix B, we utilize a unified prompt to finish ownership verification. We compare PA²Perturb with GPrompt-G to assess its effectiveness under our verification scheme.

C. Baselines

Recent advances in dataset ownership verification predominantly leverage backdoor-based methods. Given the threat model, we adopt SBA [42], GTA [43], UGBA [46], DPGBA [45], and SPEAR [47] as state-of-the-art (SOTA) baselines. To provide a more comprehensive evaluation, we further incorporate fingerprint-based methods such as GNN-Fingers [63] and GraphGuard [64] as additional baselines.

D. Evaluation Metrics

To comprehensively evaluate our ownership verification scheme, we adopt three complementary metrics that reflect both verification reliability and task integrity: verification success rate (VSR) quantifies the success rate of ownership verification the proportion of cases where both behavioral consistency and distributional alignment jointly confirm or reject model misuse with statistical confidence. Classification accuracy rate (CAR) measures the predictive performance of the model on soft nodes when the activated prompts and the CIS are integrated, which ensures that the embedding perturbations do not degrade model performance. False positive rate (FPR) is utilized to assess the reliability of results, defined as the proportion of subjects incorrectly flagged as infringer.

VII. EXPERIMENTAL RESULTS

In this section, our experimental evaluation is designed to answer the following research questions (RQs):

- **RQ1:** Can our scheme outperform SOTA dataset ownership verification methods across various GNN models?

- **RQ2:** Does PA²Perturb enable effective dataset ownership verification under various attack methods?
- **RQ3:** Whether PA²Perturb better preserves the classification accuracy rate than existing verification methods?
- **RQ4:** Whether our scheme is transferable across various GNN architectures and downstream tasks?
- **RQ5:** How do key components affect the performance of dataset ownership verification?

A. Evaluation of Verification Effectiveness (RQ1)

To answer **RQ1**, we evaluate the dataset verification performance of PA²Perturb on various real-world graph datasets, compared with SOTA baselines. The verification success rate and the false positive rate under different GNN models are presented in Table II, which provides detailed numerical results for each model. Across diverse datasets and GNN architectures, PA²Perturb consistently achieves the highest VSR with the lowest FPR. Traditional methods (e.g., SBA-Samp, SBA-Gen) fail on graph data, with VSR below 5% on large-scale datasets. Enhanced backdoor methods such as GTA, UGBA, and DPGBA perform well on standard benchmarks but degrade substantially on mixed datasets where GTA only reaches 44% VSR on Cora-F with FPR exceeding 58%. Fingerprint-based methods maintain competitive VSR yet exhibit FPR above 9%. In contrast, PA²Perturb exceeds 93% VSR with FPR below 6% on standard datasets and surpasses SPEAR by up to 18.9% on Cora-F.

Fig. 3 further evaluates the accuracy of PA²Perturb and baselines on edge-level and graph-level tasks across four datasets with GCN. SBA exhibits the lowest accuracy on both tasks, which confirms that fixed backdoor patterns cannot generalize beyond node-level classification. GTA, UGBA, DPGBA, and SPEAR achieve moderate accuracy on standard datasets but show noticeable drops on Cora-P and Cora-F, particularly on graph-level tasks where structural integrity is more critical. GNNFingers and GraphGuard perform comparably to DPGBA but remain below GPrompt-G. PA²Perturb achieves the highest accuracy on both tasks across four datasets, and the advantage is more pronounced on mixed datasets, where it outperforms the second-best method by over 3% on edge-level tasks and over 5% on graph-level tasks. These results indicate that the prompt-activated perturbation of PA²Perturb preserves strong task adaptability and does not degrade model utility across different downstream tasks.

B. Evaluation of Stealthiness and Robustness (RQ2)

To answer **RQ2**, we formally establish the notion of an adaptive adversary described in Section IV. To verify the stealthiness and robustness of PA²Perturb, we conduct evaluations under three representative attack strategies across various classical graph datasets and GNN architectures. As shown in Table III, PA²Perturb consistently achieves the highest VSR and the lowest FPR in most settings. Under the no attack setting, several advanced methods achieve reasonable VSR on standard datasets, yet their performance drops sharply under adversarial attacks. For instance, the VSR of SPEAR on Cora decreases from 95.9% to 65.3%, while PA²Perturb

TABLE II
THE RESULT (VSR|FPR)(%) ACROSS DATASETS AND GNNs. THE BEST RESULT IN EACH ROW IS HIGHLIGHTED IN BOLD.

Datasets	GNN Models	SBA-Samp		SBA-Gen		GTA		UGBA		DPGBA		SPEAR		GNNFingers		GraphGuard		GPrompt-G		Ours	
Cora	GCN	33.2	32.1	44.2	27.8	91.4	17.3	96.0	12.2	96.7	9.7	97.0	8.3	96.3	11.5	96.5	11.1	95.9	7.9	97.2	4.3
	GraphSAGE	35.7	34.5	44.7	25.6	90.7	19.8	93.5	10.8	95.3	11.2	95.8	9.1	93.8	10.2	94.0	9.9	94.8	8.7	96.1	5.1
	GAT	37.2	31.7	46.1	28.9	92.3	16.2	96.7	14.1	97.2	8.5	97.5	7.6	97.0	13.3	97.1	12.9	96.3	9.4	97.9	4.7
	GIN	36.5	33.8	45.7	26.3	91.9	18.9	95.7	11.5	94.3	10.7	95.4	9.8	96.0	10.8	96.1	10.4	96.1	7.2	96.9	5.6
Citeseer	GCN	29.4	31.9	40.1	28.2	85.7	16.7	93.2	13.4	94.4	11.8	94.6	8.1	93.5	12.7	93.7	12.3	94.2	9.3	95.3	4.9
	GraphSAGE	29.8	33.6	39.7	27.1	86.3	18.5	92.7	10.2	93.8	9.3	94.0	7.4	93.0	9.6	93.2	9.2	93.9	8.6	94.9	5.2
	GAT	30.7	30.4	41.5	29.7	89.2	15.9	95.1	14.7	95.7	10.1	96.0	9.6	95.4	14.0	95.5	13.6	95.3	7.8	96.1	4.4
	GIN	30.2	32.7	40.8	25.8	88.6	17.4	94.3	12.9	94.6	11.5	95.1	8.9	94.6	12.2	94.8	11.8	94.2	9.1	95.8	5.8
Pubmed	GCN	30.2	34.2	31.2	29.4	89.1	19.3	92.6	11.8	93.7	10.4	93.8	7.7	92.9	11.1	93.0	10.7	91.3	8.5	94.2	5.3
	GraphSAGE	30.9	31.1	29.7	27.9	88.5	16.8	90.3	13.7	91.7	11.9	92.0	9.3	90.6	13.0	90.8	12.6	90.5	7.6	92.4	4.6
	GAT	32.7	33.5	32.5	26.7	89.7	18.1	93.4	10.5	93.1	9.8	93.3	8.4	93.7	9.9	93.8	9.5	92.9	9.7	93.6	5.1
	GIN	31.6	30.8	30.2	28.3	87.2	15.6	91.8	14.2	92.4	10.6	92.5	7.2	92.1	13.5	92.2	13.1	92.7	8.1	93.1	4.8
Flickr	GCN	3.2	92.3	2.7	93.2	83.2	16.9	91.7	10.3	95.1	11.2	95.9	8.6	92.0	9.7	92.2	9.3	95.7	7.4	97.1	5.4
	GraphSAGE	1.8	94.1	1.4	91.9	80.2	18.7	90.2	14.6	92.3	9.7	93.7	7.9	90.5	14.0	90.7	13.6	94.2	9.5	96.8	4.2
	GAT	4.1	91.7	3.9	94.5	88.4	15.2	92.3	12.8	96.7	10.8	97.2	9.1	92.6	12.1	92.8	11.7	96.9	8.3	98.1	5.7
	GIN	2.6	93.5	2.1	90.7	85.7	17.8	89.4	11.4	95.9	8.9	96.3	7.5	89.7	10.8	89.9	10.4	96.1	9.9	97.3	4.5
Cora-P	GCN	3.7	90.8	4.3	92.4	55.3	45.1	63.4	32.3	72.5	28.5	77.5	26.8	63.7	31.6	63.9	31.2	82.4	17.5	87.3	7.5
	GraphSAGE	0.9	92.6	0.7	93.7	58.2	44.5	65.6	31.7	73.6	28.2	78.2	27.1	65.9	31.0	66.1	30.6	83.2	16.4	88.1	6.9
	GAT	4.5	91.2	3.1	91.1	51.8	46.7	61.2	33.5	71.3	29.4	76.8	26.2	61.5	32.8	61.7	32.4	81.7	18.2	86.6	8.1
	GIN	1.3	94.4	1.6	94.8	54.6	45.9	64.4	32.9	72.9	28.8	77.9	27.5	64.7	32.2	64.9	31.8	82.9	17.8	87.8	7.3
Cora-F	GCN	2.9	93.8	2.4	90.5	44.2	58.5	50.7	41.8	57.2	37.1	61.2	34.8	51.0	41.1	51.2	40.7	65.9	22.8	80.1	8.4
	GraphSAGE	0.5	90.3	0.3	92.1	46.6	57.9	52.5	40.2	58.9	36.7	62.6	35.2	52.8	39.5	53.0	39.1	67.6	21.3	81.7	8.9
	GAT	3.6	92.9	4.0	93.4	41.4	60.7	49.0	43.6	57.0	38.2	61.4	35.4	49.3	42.9	49.5	42.5	66.2	23.7	80.6	8.5
	GIN	1.1	91.6	1.0	91.3	43.7	59.7	51.5	42.8	58.3	37.4	62.3	36.4	51.8	42.1	52.0	41.7	67.1	23.1	81.4	9.5

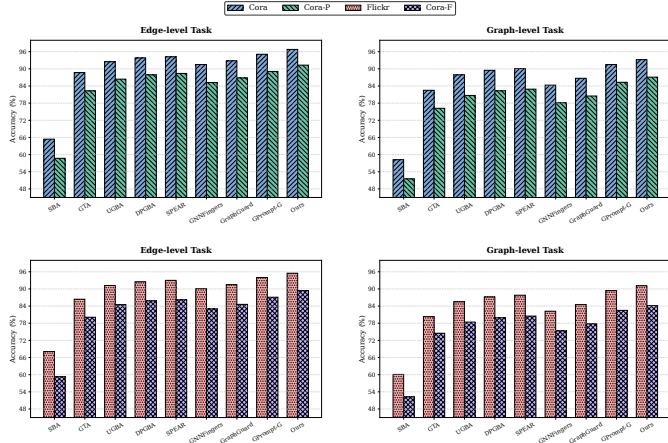


Fig. 3. The accuracy (%) on edge-level and graph-level task across different datasets with GCN.

only decreases from 97.2% to 91.2%. On mixed datasets, the gap widens further: under RIGBD on Cora-F, the majority of baselines fall below 47% VSR, whereas PA²Perturb retains 70.4%.

We further investigate the robustness of PA²Perturb under varying pruning ratios from 0% to 80%. As shown in Fig. 4, the verification accuracy of most baselines decreases rapidly as the pruning ratio increases. On Cora, methods such as GTA and SBA drop below 30% accuracy at 80% pruning, while SPEAR and GPrompt-G degrade to around 45%. In contrast, PA²Perturb maintains above 89% accuracy even at the highest pruning ratio, with a total decrease of less than 10% across the full range. Similar trends are observed on Cora-P, Flickr, and Cora-F, where PA²Perturb consistently remains the top-performing method with the slowest degradation curve. These results confirm that the perturbations injected by PA²Perturb are deeply embedded in stable graph regions and cannot be easily removed through structural pruning.

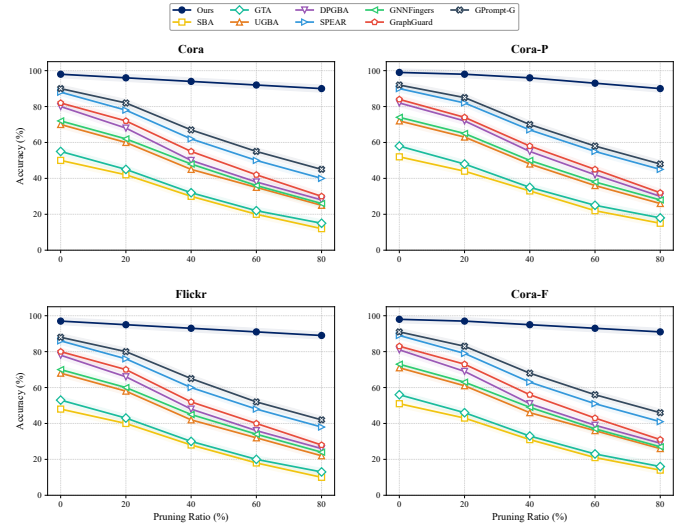


Fig. 4. Robustness comparison of our scheme and baselines under Prune attacks on different datasets.

C. Evaluation of Harmlessness (RQ3)

To answer RQ3, we conduct a comparative analysis of our proposed scheme and various baselines. The comparison is based on node classification accuracy across popular datasets and GNN models. As shown in Table IV, backdoor-based methods (e.g., SBA, GTA, UGBA, DPGBA) introduce noticeable accuracy degradation compared with the clean model, with drops of 1% to 4% on standard datasets. SPEAR mitigates this degradation but still falls slightly below the clean baseline. In contrast, PA²Perturb not only preserves but even slightly improves CAR over the clean model in most settings. For example, on Cora with GAT, the clean model achieves 86.4% while PA²Perturb achieves 87.1%. This improvement is because our scheme injects perturbations only on hard nodes and utilizes the activated prompt to correct their predictions, which

TABLE III

THE RESULT (VSR|FPR)(%) UNDER DIFFERENT ATTACK METHODS. NONE MEANS NO ATTACK IS PERFORMED. THE BEST RESULT IN EACH ROW IS HIGHLIGHTED IN BOLD.

Datasets	Attack Methods	SBA-Samp		SBA-Gen		GTA		UGBA		DPGBA		SPEAR		GNNFingers		GraphGuard		GPrompt-G		Ours	
Cora	None	33.2	32.1	44.2	27.8	91.4	17.3	96.0	12.2	96.7	9.7	95.9	8.3	95.0	12.5	95.8	11.8	96.0	7.9	97.2	4.3
	Prune	16.8	33.7	19.6	29.1	15.2	19.3	90.2	12.9	91.6	10.8	91.3	9.6	35.2	13.8	88.5	10.5	97.0	6.8	98.2	4.9
	OD	14.7	34.5	14.3	27.8	17.5	18.8	26.8	13.6	93.2	10.5	94.1	8.7	32.8	14.2	90.3	10.1	95.5	7.1	96.3	5.4
	RIGBD	11.7	33.9	13.5	28.5	18.3	19.2	27.5	13.2	32.1	10.9	89.3	9.8	65.3	12.9	85.7	9.8	90.0	7.5	91.2	5.1
Citeseer	None	29.4	31.9	40.1	28.2	85.7	16.7	93.2	13.4	94.4	11.8	94.2	8.1	92.5	14.2	93.8	13.5	94.5	9.3	95.3	4.9
	Prune	14.2	34.2	17.6	29.4	10.3	19.1	89.7	13.1	90.6	10.9	89.7	9.3	30.5	15.1	85.8	12.8	92.8	7.9	93.5	5.0
	OD	10.3	33.6	11.7	28.2	16.9	18.7	25.9	13.4	91.2	10.6	92.7	8.5	28.3	15.5	88.9	12.2	93.8	7.4	94.5	5.2
	RIGBD	10.9	34.0	12.8	28.8	17.6	19.0	26.4	13.5	31.8	10.8	94.2	9.1	60.7	14.8	83.5	11.5	95.0	7.2	95.8	4.8
Pubmed	None	30.2	34.2	31.2	29.4	89.1	19.3	92.6	11.8	93.7	10.4	91.3	7.7	91.8	13.7	92.5	12.9	93.0	8.5	93.9	5.3
	Prune	17.4	34.1	19.7	29.0	11.4	19.0	90.9	13.3	91.7	10.7	90.7	9.5	33.6	14.5	87.2	11.8	92.4	7.7	93.1	5.3
	OD	10.5	33.7	11.9	28.5	17.2	18.6	27.1	13.5	91.8	10.5	90.3	8.8	29.8	14.9	86.5	11.2	92.0	7.6	92.7	5.0
	RIGBD	10.1	33.6	11.4	28.7	18.0	18.9	28.3	13.7	33.6	10.6	89.7	9.6	58.9	13.8	82.1	10.5	90.4	7.3	91.1	5.2
Flickr	None	3.2	92.3	2.7	93.2	83.2	16.9	91.7	10.3	95.1	11.2	95.9	8.6	94.5	10.8	95.3	9.9	96.3	7.4	97.1	5.4
	Prune	3.5	93.1	5.2	90.2	14.1	18.1	89.3	13.2	91.2	10.8	92.1	9.3	28.7	12.5	86.3	9.8	93.6	7.2	94.3	4.9
	OD	4.2	91.7	6.5	88.9	12.7	18.7	20.5	13.8	94.8	10.4	93.7	8.6	25.3	13.1	89.5	9.2	94.4	7.4	95.1	5.1
	RIGBD	3.8	92.9	5.8	89.8	13.9	18.3	21.1	13.6	34.6	10.2	87.3	9.7	52.1	11.5	78.9	9.5	88.8	7.1	90.6	5.0
Cora-P	None	3.7	90.8	4.3	92.4	55.3	45.1	63.4	32.3	72.5	28.5	77.5	26.8	75.8	29.7	79.2	27.5	81.6	17.5	87.3	7.5
	Prune	3.8	93.5	5.9	89.1	15.9	74.1	26.8	64.2	35.9	54.3	45.1	48.9	22.3	68.5	38.7	45.2	54.5	13.8	74.8	6.8
	OD	3.5	94.2	5.2	89.7	14.3	74.8	25.1	64.9	34.2	55.1	43.7	49.6	20.8	69.2	36.5	46.1	53.4	14.5	73.5	8.1
	RIGBD	3.1	94.8	4.8	90.3	13.6	75.5	24.3	65.6	33.5	55.8	42.5	50.3	32.7	58.9	48.3	38.7	52.5	15.1	72.7	7.5
Cora-F	None	2.9	93.8	2.4	90.5	44.2	58.5	50.7	41.8	57.2	37.1	61.2	34.8	59.8	40.3	63.5	36.2	65.1	22.8	80.1	8.4
	Prune	2.8	94.1	4.8	89.8	12.8	74.7	22.9	64.8	31.6	54.9	40.7	49.7	18.5	71.2	32.4	52.8	49.4	14.7	71.2	8.7
	OD	2.5	94.7	4.3	90.5	11.6	75.3	21.5	65.5	30.2	55.6	39.3	50.3	17.2	72.1	30.8	53.5	48.0	15.3	70.7	9.8
	RIGBD	2.2	95.3	3.9	91.2	10.9	76.0	20.8	66.1	29.5	56.2	38.2	51.0	26.3	63.8	42.7	45.9	46.9	16.0	70.4	9.6

TABLE IV

CLASSIFICATION ACCURACY RATE (%) ACROSS DATASETS AND GNNs.

Datasets	GNNs	Clear	SBA	GTA	UGBA	DPGBA	SPEAR	Ours
Cora	GCN	85.8	83.6	82.9	83.5	84.1	85.2	86.3
	GraphSAGE	85.6	82.9	83.3	83.6	83.7	84.9	86.1
	GAT	86.4	84.4	83.9	83.3	83.7	85.4	87.1
	GIN	85.9	83.6	83.1	83.4	83.3	84.9	86.6
Citeseer	GCN	76.9	75.4	75.0	75.4	75.6	76.4	77.3
	GraphSAGE	76.8	75.1	75.4	75.4	75.7	76.4	77.1
	GAT	77.1	75.5	75.3	75.0	75.2	76.4	77.6
	GIN	76.5	75.3	75.0	75.2	75.2	76.0	76.9
Pubmed	GCN	86.3	85.0	85.3	85.1	85.1	85.9	86.7
	GraphSAGE	86.3	84.9	85.1	85.2	85.5	86.0	86.5
	GAT	86.5	84.2	83.8	85.4	85.1	86.0	86.9
	GIN	85.9	83.6	83.1	83.4	83.3	84.9	86.6
Flickr	GCN	50.1	46.0	45.0	45.9	46.6	48.4	51.2
	GraphSAGE	49.2	44.7	45.3	46.1	46.3	47.8	50.3
	GAT	50.0	46.8	46.6	45.7	45.9	48.1	51.4
	GIN	49.7	45.7	45.0	45.5	45.7	48.0	50.9
Cora-P	GCN	84.2	82.0	81.5	81.8	82.3	83.5	84.7
	GraphSAGE	83.9	81.3	81.7	82.0	82.1	83.2	84.4
	GAT	84.6	82.7	82.2	81.9	82.4	83.8	85.2
	GIN	84.1	81.8	81.4	81.6	81.7	83.2	84.8
Cora-F	GCN	45.4	42.3	41.5	42.0	42.7	44.4	46.1
	GraphSAGE	44.7	41.0	41.8	42.4	42.5	43.9	45.3
	GAT	45.5	43.0	42.8	41.9	42.1	44.2	46.3
	GIN	45.1	41.9	41.3	41.7	41.8	43.9	46.0

effectively acts as a regularizer that enhances the decision boundary on ambiguous samples.

To further validate the imperceptibility of the injected perturbations, we compare the reconstruction loss distributions between normal nodes and perturbed nodes. As shown in Fig. 5, the reconstruction loss of perturbed nodes falls well within the distribution of normal nodes on Cora, Cora-P, Flickr, and Cora-F. The two distributions exhibit substantial overlap, which indicates that the perturbations generated by PA²Perturb are statistically indistinguishable from normal graph structures. This result confirms that the semantic and distributional similarity constraints of the CIS generator effectively prevent the injected perturbations from becoming outliers, and there-

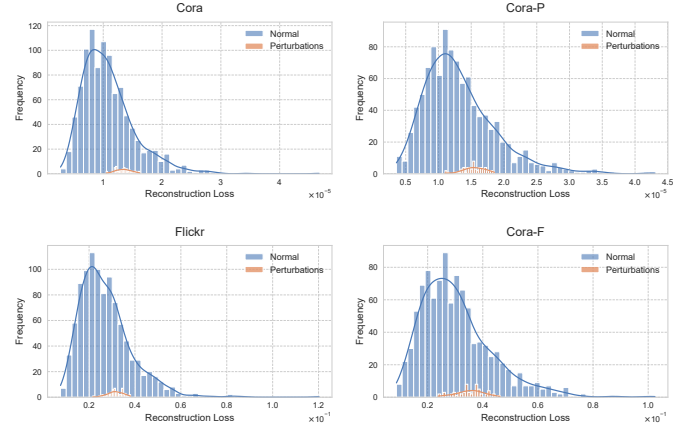


Fig. 5. Reconstruction loss distributions on node-level task across various datasets.

fore the perturbed dataset cannot be easily identified through anomaly detection methods.

D. Transferability Evaluation (RQ4)

To answer **RQ4**, we investigate the effectiveness of PA²Perturb on rigid experimental conditions, which means that the dataset owner has minimal knowledge about suspected models. In this process, we keep the activated prompts obtained from the surrogate model unchanged and directly feed them into the query as part of the input for detecting suspected models. As shown in Table V, our PA²Perturb exhibits excellent transferability, with an average VSR of over 88.7%. Specifically, the PA²Perturb shows exceptional transferability when both surrogate and suspected models share similar architectures. For example, our PA²Perturb can effectively transfer between GCN and GraphSAGE, which may be attributed to the fact that both models utilize highly similar aggregation mechanisms. Interestingly, GAT performed

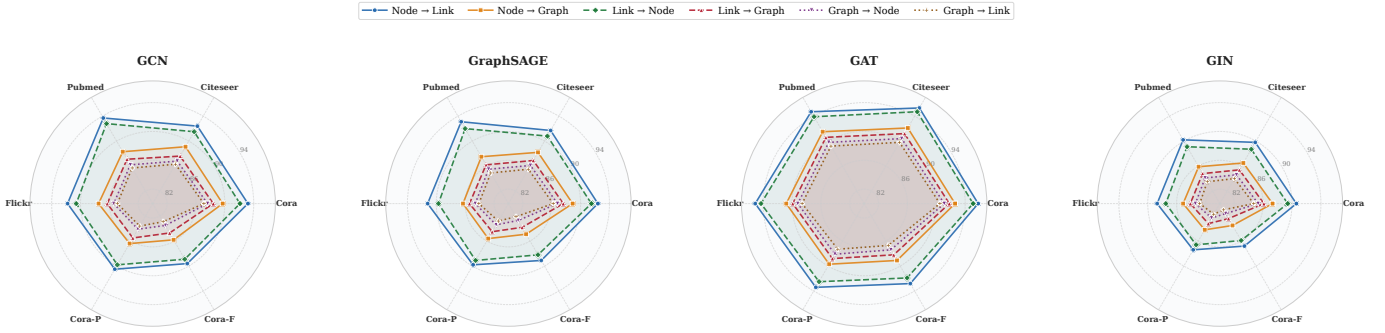


Fig. 6. Transferability of PA²Perturb across different task levels with fixed GNN architectures. Each subplot corresponds to a specific GNN, and the six axes represent different datasets.

TABLE V
TRANSFERABILITY OF PA²PERTURB ACROSS DIFFERENT GNN ARCHITECTURES AND DATASETS.

Surrogate Models	Suspected Models	Cora	Citeseer	Pubmed	Flickr	Cora-P	Cora-F
GCN	GraphSAGE	92.4	91.2	93.1	90.2	90.8	88.6
	GAT	88.7	89.5	87.2	90.3	86.1	85.7
	GIN	85.9	85.7	83.9	84.1	83.2	82.4
GraphSAGE	GCN	91.0	91.9	92.2	90.1	89.7	88.3
	GAT	89.1	89.8	86.9	90.2	85.8	86.1
	GIN	84.1	85.4	85.8	83.7	82.9	81.5
GAT	GCN	95.3	95.1	93.2	94.8	92.4	92.1
	GraphSAGE	94.7	95.9	93.7	96.1	92.0	91.7
	GIN	91.6	92.7	92.9	91.6	90.1	89.3
GIN	GCN	87.5	88.3	88.7	86.9	85.8	84.6
	GraphSAGE	89.2	89.0	86.0	87.4	85.3	84.9
	GAT	86.1	87.5	85.2	87.8	83.7	83.2

exceptionally well compared to other surrogate models, with an average VSR improvement of 2.5% and was the only model with a VSR exceeding 93.7%. Furthermore, despite building on the MLP encoder architecture, the PA²Perturb based on GIN can still effectively adapt to encoder differences caused by nonlinear transformations.

Fig. 6 presents the cross-task transferability of PA²Perturb with fixed GNN architectures. Each radar chart depicts six transfer directions among node-level, link-level, and graph-level tasks. We observe that transfers between node-level and link-level tasks consistently achieve the highest VSR, which exceeds 90% under GAT and remains above 87% under GIN. Transfers that involve graph-level tasks show relatively lower but still competitive performance, as graph-level tasks require holistic structural reasoning that differs from local node or edge representations. Among the four GNN architectures, GAT yields the strongest cross-task transferability with most directions above 88%, while GIN exhibits the lowest. These results demonstrate that PA²Perturb can effectively transfer perturbation signals across different task levels, which validates the task-agnostic nature of our activated prompt design.

E. Ablation Studies (RQ5)

To answer **RQ5**, we conduct an ablation study to examine the contributions of key components in PA²Perturb. To assess the effectiveness of region selection, we design a PA²Perturb variant named PA²Perturb-R, which randomly selects regions

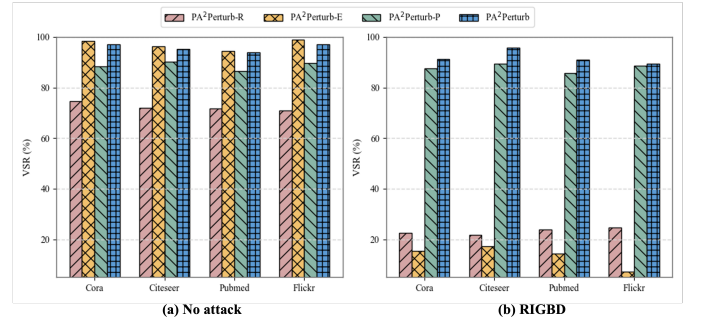


Fig. 7. Ablation studies about PA²Perturb and its variants. PA²Perturb maintains superior and stable verification accuracy under both none attack and RIGBD settings, which indicates that all key components are essential for robust performance.

from the original graph as targets for perturbation injection. To evaluate the contribution of the cue-sensitive imperceptible perturbation in improving stealthiness, we remove both the distribution similarity and semantic similarity constraints from the CIS generator and referred to this variant as PA²Perturb-E. To verify the effectiveness of our activated prompt design in enhancing verification accuracy, we replace it with a single unified prompt module and name the variant PA²Perturb-P.

We compare these three variants with the PA²Perturb under both the none attack and RIGBD settings on GCN architecture. Fig. 7 presents the verification accuracy results of the four methods across various popular datasets. To ensure a fair comparison, the hyper parameters of each variant are tuned based on performance on the validation set. Under the no attack setting, PA²Perturb-R shows a significant VSR decline of approximately 20% compared with PA²Perturb, as randomly selected regions are often structurally unstable and lead to easily identifiable perturbations. PA²Perturb-E achieves slightly higher VSR than PA²Perturb in some cases because unconstrained perturbations are more distinguishable, which increases the output similarity between legal and suspected models. PA²Perturb-P applies a unified prompt to nodes regardless of their hardness, which limits its ability to emphasize hard nodes and results in slightly reduced VSR. Under the RIGBD attack, PA²Perturb-R drops further because the attack weakens the already limited activation effect. PA²Perturb-E suffers the most severe degradation, as its more conspicuous

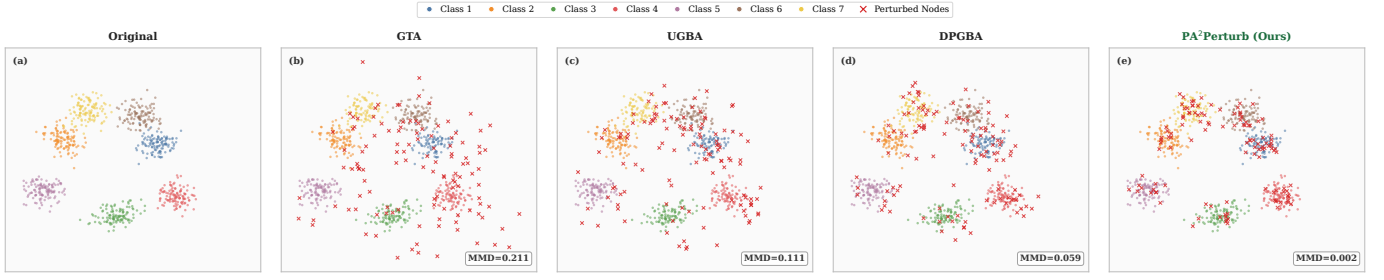


Fig. 8. t-SNE visualization of node embeddings on Cora before and after perturbation injection by different methods. Red crosses denote perturbed nodes. The MMD value quantifies the distributional shift between original and perturbed embeddings. PA²Perturb achieves the lowest MMD, indicating that its perturbations are nearly imperceptible in the embedding space.

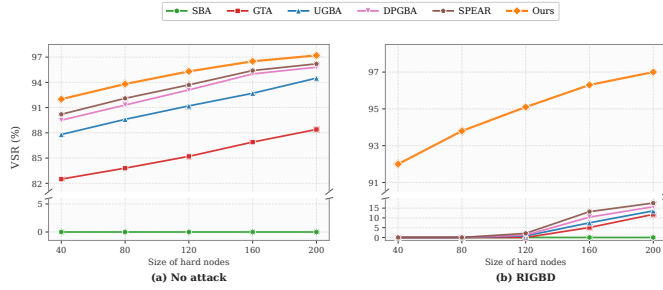


Fig. 9. Comparison of verification success rates (%) of baseline methods and PA²Perturb with different hard node sizes.

perturbations are easier for the attacker to detect and remove. Once these perturbations are disrupted, the prompt loses its amplification target and verification accuracy collapses below 25%. PA²Perturb maintains the highest and most stable VSR under both settings, which confirms that each component contributes to the overall robustness of the verification scheme.

Fig. 8 visualizes the t-SNE embeddings on Cora before and after perturbation injection by different methods. In the original graph, seven classes form well-separated clusters. GTA causes the most severe distributional shift, as its fixed trigger pattern forces perturbed nodes to deviate significantly from their original clusters. UGBA reduces the shift through adaptive trigger generation, yet many perturbed nodes still scatter outside the cluster boundaries. DPGBA further mitigates the displacement with distribution-aware constraints, but noticeable drift remains visible. PA²Perturb achieves the lowest MMD among the four methods, where the perturbed nodes are nearly indistinguishable from normal nodes and remain tightly within their respective class clusters. This result confirms that the semantic and distributional similarity constraints of the CIS generator effectively align the injected perturbations with the original graph structure, which prevents identification through embedding-level anomaly detection.

F. Perturbation Budget Analysis

We conduct experiments to investigate the verification performance of PA²Perturb under varying numbers of hard nodes. Specifically, we set the number of selected nodes to {40, 80, 120, 160, 200} and compared PA²Perturb with other methods under both none attack and RIGBD strategies on Citeseer



Fig. 10. Hyper-parameter sensitivity analysis. PA²Perturb achieves optimal verification accuracy when both β and γ are greater than 1, and the two parameters are adjusted proportionally to balance verification performance and perturbation stealthiness.

dataset. Fig. 9 presents the results of various methods across different perturbation budgets. We report only the verification success rate, as no significant changes in classification accuracy rate are observed across various baselines and PA²Perturb. From Fig. 9, we can observe the following:

Under the none attack setting, our scheme consistently maintains the highest verification accuracy across different hard node scales. As the number of nodes increases from 40 to 200, the VSR steadily improves from 91.0% to approximately 97.8%, which demonstrates significant scalability and stability. In contrast, although baseline methods show a certain upward trend, their overall performance remains significantly lower than that of PA²Perturb. The traditional method SBA performs poorly, with VSR consistently below 5.0%, which shows that it is insensitive to node quantity and lacks sufficient verification capability.

Under the RIGBD attack, our scheme still demonstrates superior performance, with VSR increasing from 92.0% to 97.0%, and it significantly outperforms various other methods. Moreover, the growth curve has been consistently smooth and stable. In contrast, the other methods almost completely fail under low node scales and show only slight improvement as the number of nodes increases. Among them, SPEAR performs relatively better but still lags behind by a noticeable margin. These results indicate that PA²Perturb retains excellent resilience even under advanced attack strategies.

G. Hyper-Parameter Sensitivity Analysis

We further investigate how the hyper parameters β and γ affect the performance of PA²Perturb, where β controls the

enhancement of verification performance and γ determines the weight of the distribution similarity and semantic similarity constraints. To explore the influence of β and γ , we fix the surrogate model as GCN and conduct experiments on the Citeseer dataset, where the values of β and γ are varied in $\{0.01, 0.1, 1, 10, 100\}$. We report the verification accuracy under both none attack and RIGBD attack settings, as illustrated in the heatmaps shown in Fig. 10. We make the following observations: 1) in the absence of attack strategies, verification accuracy improves as β increases, whereas higher values of γ result in degraded accuracy; 2) After deploying attack methods, verification accuracy decreases when β exceeds 1. In contrast, increasing γ leads to improved accuracy. To enhance the stealthiness of perturbations and simultaneously preserve high verification performance, we recommend setting both β and γ to values greater than 1. Moreover, γ should be increased proportionally as β increases, so that tuning remains synchronized between different parameter pairs.

VIII. CONCLUSION

In this paper, we propose a novel scheme for graph dataset ownership verification from the perspective of utility-preserving injection. The following conclusions can be drawn from our extensive experiments across real-world scenarios: 1) the cue-sensitive perturbation can be reliably activated and amplified by dual prompts, which allows for ownership signals to be adaptively revealed without harming model performance; 2) the proposed scheme demonstrates remarkable robustness under adaptive adversarial attacks due to the semantic and distributional alignment of the injected perturbations; 3) our PA²Perturb demonstrates powerful transferability across different datasets and GNN architectures, which ensures consistent verification performance. Overall, our empirical experiments demonstrate that PA²Perturb outperforms current SOTA methods against the usage of unauthorized graph-structured datasets. We reveal that models fine-tuned on the same dataset share the underlying perturbation affinities. In the near future, we will further investigate ownership verification of heterogeneous graph datasets and analyze how different entity categories impact the verification performance.

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